

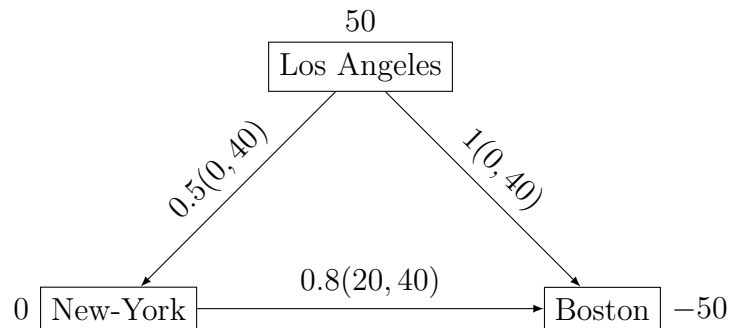
Transshipment problem

Michel Bierlaire

Solution of the practice quiz

1. The network corresponding to the problem is represented below. Each city is represented by a node. Each pair of cities is connected with an arc, oriented according to the definition of costs. The supply/demand for each city is

- Boston: $200 - 250 = -50$,
- New-York: $250 - 250 = 0$,
- Los Angeles: $300 - 250 = 50$.



2. The transshipment problem is defined as follows.

$$\min_{x \in \mathbb{R}^3} 0.5x_{LN} + x_{LB} + 0.8x_{NB}$$

where the decision variables are the flow on each arc, subject to the flow conservation constraints for each node:

$$\begin{aligned}x_{LN} + x_{LB} &= 50, \\x_{NB} - x_{LN} &= 0, \\-x_{LB} - x_{NB} &= -50,\end{aligned}$$

and subject to the capacity constraints for each arc:

$$\begin{aligned} 0 &\leq x_{LN} \leq 40, \\ 0 &\leq x_{LB} \leq 40, \\ 20 &\leq x_{NB} \leq 40. \end{aligned}$$

3. To obtain the standard form, we first set the lower bounds to zero by a change of variables $x'_{ij} = x_{ij} - \ell_{ij}$. We define

$$\begin{aligned} x'_{LN} &= x_{LN}, \\ x'_{LB} &= x_{LB}, \\ x'_{NB} &= x_{NB} - 20. \end{aligned}$$

The problem becomes

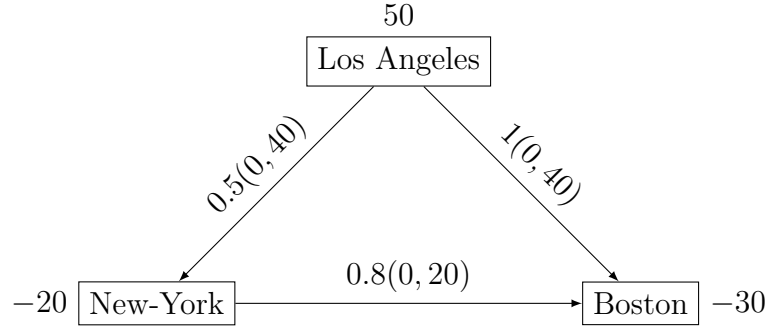
$$\min_{x \in \mathbb{R}^3} 0.5x'_{LN} + x'_{LB} + 0.8x'_{NB} + 16$$

subject to the flow conservation constraints for each node:

$$\begin{aligned} x'_{LN} + x'_{LB} &= 50, \\ x'_{NB} - x'_{LN} &= -20, \\ -x'_{LB} - x'_{NB} &= -30, \end{aligned}$$

$$\begin{aligned} 0 &\leq x'_{LN} \leq 40, \\ 0 &\leq x'_{LB} \leq 40, \\ 0 &\leq x'_{NB} \leq 20. \end{aligned}$$

Not that the constant term in the objective function (16) can be ignored.



To create the network corresponding to the standard form, we need to consider all arcs (i, j) with an upper bound on the flow (all three of them in this example). For each of them, we introduce a new node associated with a supply defined by the upper bound of the arc. Then we replace the original arc by

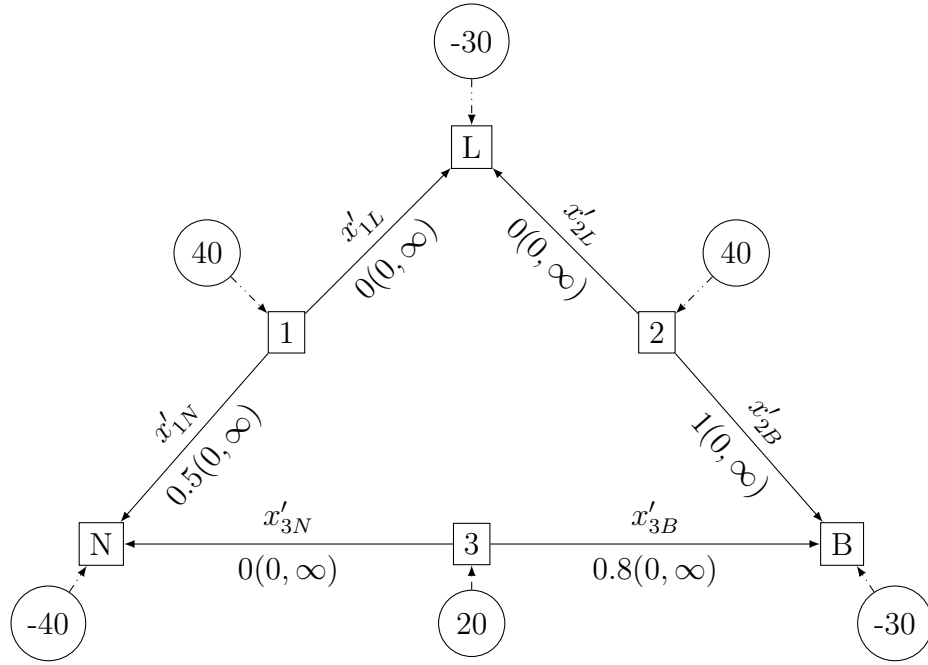
- an arc from the new node k to the downstream node j of the original arc, with flow x_{kj} and cost $c_{kj} = c_{ij}$ and
- an arc from the new node k to the upstream node i of the original arc, with flow y_{ki} and cost $c_{ki} = 0$.

We also have to recalculate the supply/demand of the original upstream nodes using the formula

$$s'_i = s_i - \sum_{(i,j) \in \mathcal{A}} u_{ij}.$$

- $s'_L = s_L - u_{LN} - u_{LB} = 50 - 40 - 40 = -30$,
- $s'_N = s_N - u_{NB} = -20 - 20 = -40$,
- $s'_B = s_B = -30$.

We obtain the following network, where node 1 is associated with arc (L, N) , node 2 is associated with arc (L, B) and node 3 is associated with arc (N, B) .



The corresponding linear optimization problem is:

$$\min x'_{2B} + 0.8x'_{3B} + 0.5x'_{1N}$$

subject to the flow conservation constraints

$$\begin{aligned} -x'_{1L} - x'_{2L} &= -30 & (\text{node } L), \\ x'_{1L} + x'_{1N} &= 40 & (\text{node } 1), \\ -x'_{1N} - x'_{3N} &= -40 & (\text{node } N), \\ x'_{3N} + x'_{3B} &= 20 & (\text{node } 3), \\ -x'_{2B} - x'_{3B} &= -30 & (\text{node } B), \\ x'_{2L} + x'_{2B} &= 40 & (\text{node } 2), \end{aligned}$$

and the non negativity constraints on each flow

$$x'_{1L}, x'_{1N}, x'_{2L}, x'_{2B}, x'_{3N}, x'_{3B} \geq 0.$$

4. If we order the variables as $x'_{1L}, x'_{1N}, x'_{2L}, x'_{2B}, x'_{3N}, x'_{3B}$, the problem is

$$\min_{x' \in \mathbb{R}^n} c^T x'$$

subject to

$$\begin{aligned} Ax' &= b, \\ x' &\geq 0, \end{aligned}$$

where $m = 6$, $n = 6$,

$$A = \begin{pmatrix} -1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} -30 \\ 40 \\ -40 \\ 20 \\ -30 \\ 40 \end{pmatrix}, \quad c = \begin{pmatrix} 0 \\ 0.5 \\ 0 \\ 1 \\ 0 \\ 0.8 \end{pmatrix}.$$

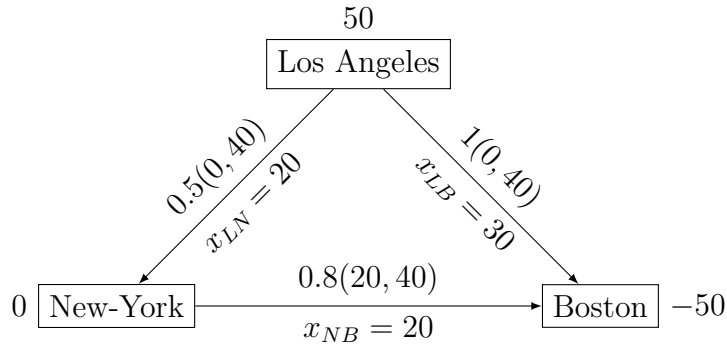
The optimal solution is $x'_{1L} = 20$, $x'_{1N} = 20$, $x'_{2L} = 10$, $x'_{2B} = 30$, $x'_{3N} = 20$, $x'_{3B} = 0$. In the network without slack variables, it means

$$\begin{aligned} x'_{LN} &= 20, \\ x'_{LB} &= 30, \\ x'_{NB} &= 0. \end{aligned}$$

And in the original variables, it means

$$\begin{aligned} x_{LN} &= 20, \\ x_{LB} &= 30, \\ x_{NB} &= 20. \end{aligned}$$

The optimal flows on the original network are represented in the figure below.



5. The matrix A is

$$A = \begin{pmatrix} -1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

It has the property that each column contains exactly two non zero entries: 1 and -1. This property is actually shared with any transshipment problem. It means that if we sum all the rows of the matrix, we obtain the null vector. Therefore, the rows are linearly dependent, and the matrix is not of full rank. Its rank happens to be 5, not 6. There is actually a redundant constraint. Indeed, as the sum of the divergences of all nodes must be zero, one of the flow conservation constraints will be automatically verified if the others are. Consequently, one constraint of a transshipment problem can be removed before solving it.